

## Research Assignment 02

## The Language of Business Analytics

## Section 01

## Foundational Concepts of Analytics

## (1) Business Analytics

- The role of business analytics means looking at business information to understand what is happening, why is it happening and what might happen next
- The information gathered is used to solve problems.

Example:

- Alisa's home loan sales have decreased
- Business analytics would help check how much sales have gone down, when did they start decreasing and/or what could improve sales?

## (2) Business Intelligence (BI)

- Means collecting and organising business information so that managers and employees can easily see what is happening in the business.

Example:

- Think of a business intelligence as a dashboard of a car
- A dash board shows a speedometer, fuel gauge, lights and ensures all the computer parts work together effectively.

## (3) Data Analytics

- Data Analytics examine data to find patterns, relationships and useful information. Data can come from anywhere the bank, social media, mobile apps and/or customer apps.

Example:

- 50 customers walk into Kopy Kreme everyday. A data analyst might examine the data and discover that: Customers aged 10-25 years old buy Sprinkled donuts every 2 hours. That would be a pattern found in the data.

→ This information can be used to better understand its customers better

#### ④ Data Driven Decision Making

→ This means using evidence and information to make decisions instead of relying only on guesses, assumptions or feelings.

Example:

→ If the bank manager says, "I think customers don't need credit anymore." That's an opinion or assumptions.

→ Instead look at actual applications, between a certain age, with a high approval rate and a salary bracket of R 250k. Now you can make a decision based on evidence.

\* Data tells you what is happening. Analytics help you understand it. Business intelligence helps you see it. Data driven decision making helps you act on it.

#### ⑤ Big Data

→ Big Data analytics is the process of examining very huge, complex and rapidly generated amounts of data to discover trends, patterns and useful insights. The data can come from many sources.

Example:

→ Imagine a bank has millions of transactions every month. It would be extremely difficult for a human being to look at every transaction manually. Like ATM withdrawals, loan applications, customer complaints etc.

→ Certain transaction may have unusual patterns, this information can help the bank identify transactions that may be fraudulent which will require further investigation.

#### ⑥ Descriptive Analytics

→ It involves analysing historical data to understand what has already happened. It summarises past or current data using things such as reports, charts, statistics.

and dashboards.

Example:

↳ Lets say a bank wanted to track the performance of each branch for a certain period or month. It could analyse

| Measure          | Result          |                                                                                                                             |
|------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------|
| Home Loans       | 70              | * This info explains what happened during that month. It does not explain why it happened or predict what will happen next. |
| Credit cards     | 126             |                                                                                                                             |
| Personal loans   | 90              |                                                                                                                             |
| New customers    | 400             |                                                                                                                             |
| Federal policies | 890             |                                                                                                                             |
| [Profit]         | = R 3.1 million |                                                                                                                             |

### ⑨ Diagnostic Analytics

↳ Diagnostic analytics goes beyond describing what happened. It examines data to identify the causes or reason or factors that may explain why something happened.

Example:

\* Lets say credit card sales in March decreased by 25%. diagnostic analytics can answer 'why did sales decrease?', 'why customers are leaving', 'what competitors <sup>are</sup> doing'?

↳ The purpose is to investigate the possible reasons behind the result.

### ⑩ Predictive Analytics

↳ It uses historical data, statistical techniques and ~~and~~ in many cases, machine learning models to estimate what might happen in the future.

Example:

- ↳ A company can use information about previous clients and their behavior to identify customers/clients who may be more likely to apply for a home loan or credit card in the future.
- ↳ It produces an estimate based on available data.

Predictive = What might happen?

→ Simple way to remember what it means.

## ⑨ Prescriptive Analytics

→ It is the use of data to recommend what action should be taken. It also uses data, predictions, rules and sometimes AI or mathematical optimisation to suggest an appropriate action.

Example:

→ Imagine a bank wants to reduce the number of customers who leave the bank. The bank could consider different analytics - descriptive, diagnostic, predictive analysis.

→ It helps reduce costs, manage risks, improve sales and profitability.

## ⑩ Data Literacy

→ Data literacy is the ability to understand, interpret, communicate and use information from data.

→ We can also define it as the ability to read, write and communicate data in context, including understanding data sources and analytical methods, and using that understanding to make decisions.

Example:

→ [Monthly Sales]

January R 90 000

March R 100 000

May R 100 000

February R 120 000

April R 80 000

June R 150 000

→ From this data literacy should be able to look at this info and understand that - "Sales were okay in January were okay and increased dramatically in April." It could start asking "why sales dropped in April? Is it unusual?" and make a decision based on the gathered data.

## ⑪ Structured Data

→ Structured data that is organised in a predictable format, usually in rows and columns. We can also describe it as information with a predefined format that fits

neatly into rows and columns, such as information stored in spreadsheets and relational databases.

- Because the data is organised, it becomes easier for people and computer system to store, analyse, and use the data for reports and ~~data~~ business intelligence.

**Structured Data = organised data that fits neatly into a table**

## (12) Unstructured Data

- Unstructured data is information that does not follow a fixed format or fit neatly into rows and columns.

- It does not have a predefined schema and can include text, images, video and audio.

Example:

- A client complains - "The client has been trying to make a transfer on her mobile app since the morning, but the transaction keeps failing. I am extremely frustrated because I need to make this transfer urgently."

This is unstructured data.

## (13) Data Warehouse.

- A data warehouse is a central place where large amounts of data from different sources are collected, stored and organised so that it can be analysed.

Example:

- Supermarkets collect client information, transaction or a daily their spending patterns etc. The information can be brought together and stored in a data warehouse so that the ~~bank~~ supermarket can analyse its overall business performance.

## (14) Data Mining

- It is the process of examining large amounts of data to find useful patterns or trends.

Example: A bank analyses customer data and discover that customers who have a home loan are also more likely to have insurance.

→ Data mining finds useful patterns in data that were not so obvious at first.

### ⑮ Data Visualization

→ Data visualization is the process translating raw, complex numbers and unstructured information into visual formats - such as charts, graphs or diagrams.

Example:

↳ We used to use phone books and it contained lots of addresses. Imagine going through that just to access one address.

↳ Now we have apps like Google maps which does the same exact where you access all roads, shortcuts, places to eat visible in one app.

### ⑯ Dashboard.

→ It is a single, organized link that consolidates key metrics and visualizations in one place, giving users a high level overview of performance in real time.

Example:

↳ A business dashboard functions just like the dashboard in your car. All the complex parts work together (in the background) making your car engine work, speedometer, fuel gauge,

### ⑰ Metrics

↳ A metric is any measurable, standard unit measurement used to track and assess a specific process or activity. Simply put, it is a number that counts or calculates something over time.

Example:

→ In a business, this would include total website visitors, number of support calls received or monthly revenue.

### ⑮ Key Performance Indicators (KPI)

→ A KPI is a special, high priority metric that directly measures progress toward a critical, strategic objective, not every metric is a KPI.

#### Difference between a Metric & a KPI

| Metric                                                                        | KPI                                                                                                                                        |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| → Tell you what is happening e.g. "Our website had 8000 visitors this month." | → Tell you how well you are performing against a set goal. E.g. The target to resolve customer queries under 2 minutes is (Target: 7.85%). |

### ⑯ Benchmark

→ A benchmark is a standardized point of reference - a baseline standard or best practice - used to compare and evaluate performance, quality or progress over time.

## (20) Segmentation

→ Segmentation is the process of dividing a large, diverse group or dataset into smaller, distinct sub groups (segments) based on shared characteristics, having behaviors or needs.

Example:

→ In business for instance, rather than treating all millions of customers as one identical crowd, you segment (separate) them, (by race, age, spending habits, location or needs) so you can tailor your products, messages or services specifically to each group.

## (21) Conversion Rate.

→ It is the percentage of total visitors who complete a desired action - such as signing up for a newsletter, making a purchase or downloading a guide.

→ It shows how effective your website, store layout or sales process is at convincing casual browsers to become active buyers.

Example:

→ Imagine 150 people walk into a shoe store, but only 20 of them actually buy a pair of shoes before leaving. The conversion rate is 30%.

## (22) Customer Acquisition Cost (CAC)

→ It is the total amount of money spent on sales and marketing to gain a single new paying customer.

Example:

→ Imagine as a business you spend R36 000 on social media advertisements, flyers and sales commissions in one month. If these efforts bring in 10 new clients/customers, your CAC is R3600 per person ( $R36\,000 \div 10$ ).

→ The CAC tells or informs you if your marketing spend is efficient. If getting a customer costs more than the money the new customer will bring into my business, my current business model is losing money.

### (23) Customer Lifetime Value (CLV or LTV)

→ Customer Lifetime Value estimates the total profit a single customer will generate for your business over the entire duration of a relationship.

Example:

→ A customer buys coffee from your store every day for 2 years at R35 and remains loyal to you. Over this period the client would have spent R12775 a year, totalling a lifetime value of: R25550

\* Thumb rule is the CLV should be higher than CAC - at least 3 times higher

### (24) Churn Rate

→ Churn Rate also known as the attrition rate, is the percentage of customers or subscribers who stop buying from my shop or cancel their subscriptions over a certain period of time.

Example:

→ Let's say you own a club and at the beginning of the month you have 100 customers who buy beer, but by the end of the month 20 clients don't pitch up because of a new spot around. Your churn rate would be 20%.

→ It helps measure customer retention and product satisfaction. The higher the churn rate = more expenditure to acquire more customers.

### (25) Customer Retention Rate.

→ Is the percentage of existing customers who remain loyal paying clients over a specific period, excluding any new customers acquired during that time.

Example:

→ Let's say at beginning of the year you have 120 new gym members. Then during the year 20 members cancel their memberships. Your retention rate is 84% - meaning you successfully kept 100 out of the original 120 members.

- It then becomes important that for cheaper to keep existing customers than getting new ones. High retention indicates loyalty, high customer satisfaction and sustainable long term revenue.

## ②6 Net Promoter Score (NPS)

- Net Promoter Score is a standardized metric used to measure customer loyalty, satisfaction and brand advocacy. It is based on asking customers a single core question on a scale of 0 to 10: "How likely are you to recommend our company to a friend or colleague?"

- Respondents are grouped according to how they answered score (9-10), (6-8) (0-6)

Example:

→ Score 0-6 - detractors

Score 7-8 - Passives

Score 9-10 - Promoters

Calculation

$$\text{NPS} = \% \text{ of promoters} - \% \text{ of detractors}$$

## ②7 Average Order Value (AOV)

- It measures the average total amount a customer spends every time they place an order. It is calculated by dividing total profits by total number of orders

Example:

→ Lets say a grocery store makes about ₹8000 in total sales across 10 individual customer checkouts in one morning, the AOV is ₹800 per transaction ( $₹8000 \div 10$ ).

- When you increase the AOV allows a business to generate more profits from existing traffic without having to spend additional funds attracting new customers.

## ②8 Sales Funnel

- It is a visual model that shows the step by step journey a potential customer takes from first becoming aware of a brand to making a new purchase.

Example:

### → The Primary stages

Top (Awareness) → Middle (consideration) → Bottom (Purchase)  
x People discover the brand x potential buyers, evaluation x final decision

Example 1.2

- 1) Think of a restaurant window, 80 people walk past the window (awareness), 15 stop to read the poster / menu, and 5 walk in (consideration) sit down and order a meal.

### ② Marketing Campaign

A marketing campaign is a coordinated, strategic series of activities designed to promote a specific product, service or message across one or more media channels

Example:

- 1) Now with the upcoming elections, politicians just don't make as many appearances on TV, they run structured campaigns, using passing posters, door to door visits about core message to get people to vote on election day

### ③ Post campaign Analysis

A formal review conducted after a campaign ends to evaluate its overall success, measure real results against initial targets and identify lessons learned for future efforts.

Example:

- 1) After a soccer match the team reviews the game footage on Monday morning at training. They don't just check the final score, they break down the game - failures, misses, opportunities, what worked and what didn't so they can determine what needs to be improved and all

### ④ Incremental Revenue

It is the extra, additional money a company earns as a direct result of a specific action - like launching a new campaign - beyond what would have been earned anyway through normal baseline operations.

Example: A bakery normally makes 120,000 every week without running ads. To boost sales the owner runs a flyer campaign and makes 122,000 that week. The incremental profit is 12,000 ( $122,000 - 110,000$ ).

## ③② Return on investment (ROI)

↳ ROI is a financial ratio that measures how profitable an expenditure is relative to its cost. It is expressed in percentage.

Example:

Formula

$$\text{ROI} = \left( \frac{\text{Net Profit from investment}}{\text{Total cost of investment}} \right) \times 100$$

↳ If you spend R100 on supplies to build a lemonade stand and end up making R180 in total, your profit is R80. Your ROI is 80% ( $R80 : R100 \times 100$ ). This means that for every dollar you get back plus an extra 80 cents in profit.

## ③③ Return on Ad Spend (ROAS)

↳ Return on Ad Spend (ROAS) is a financial metric that measures how much gross revenue your business generates for every single dollar spent directly on advertising.

Example: Formula

$$\text{ROAS} = \frac{\text{Gross revenue from Ads}}{\text{Total Cost of Ads}}$$

↳ If you put R10 into a vending machine and the machine gives you back 9 soda plus R2 in cash. Your ROAS is 10:2 or (200%). It means for every R2 you spend/dropped into advertising, you made R20 back in revenue.

Difference between ROF and ROAS - ROI factors in all business expenses to calculate net profit. ROA strictly looks at gross revenue compared only to direct ad spend.

## ③④ Click-Through Rate (CTR)

↳ Is the percentage of people who saw an advertisement, link or email and actually clicked on it.

Example: Formula

$$\text{CTR} = \left( \frac{\text{Total Clicks}}{\text{Total impressions/ views}} \right)$$

↳ It tells you how compelling, relevant and eye catching your ad headline image or offer is to the audience seeing it.

### (35) Bounce Rate

↳ Bounce Rate is the percentage of website visitors who arrive on a webpage and leave immediately without clicking on anything else or navigating to a second page.

Example:

↳ When you walk into a clothing store, taking one step inside the doorway, realizing you're in the wrong place or don't like what you see, and immediately turning around and walking out without looking at any clothing.

### (36) Bounce Rate

↳ ~~Is the percentage of website visitors who arrive on a webpage and leave~~

### (36) A/B Testing

↳ Is an experimental method where two different versions of a single variable (e.g., webpage, email button or ad headline) are shown to two similar groups of users at the same time to see which version will do better.

Example: How it works

↳ Version 1: (Control): The existing version (e.g., a website with a Blue "Buy Now" button).

↳ Version 2: The altered version (e.g., the exact same website, but with a Green "Buy Now" button).

### (37) Lead Generation

↳ It is the process of attracting people to a business and getting them to make their intent to express interest.

Example: If you opened a sand store, if you stand inside waiting, no one knows that you exist. But if you hire to hand out flyers outside for free sample opening balloons in exchange for a phone number, I am performing lead generation. The goal is to collect as many as you can so the sales team can talk to them and share more information.

1. **Few, S. (2006).** *Information Dashboard Design: The Effective Visual Communication of Data.* O'Reilly Media. (Defines dashboard architecture, visual communication, and effective metric layout).
2. **Kaplan, R. S., & Norton, D. P. (1996).** *The Balanced Scorecard: Translating Strategy into Action.* Harvard Business School Press. (The foundational framework establishing how KPIs connect to strategic objectives).
3. **Tufte, E. R. (2001).** *The Visual Display of Quantitative Information* (2nd ed.). Graphics Press. (The seminal textbook on data visualization principles and graphical integrity).

4. **Farris, P. W., Bendure, N. T., Pfeiffer, E., & Reibstein, D. J. (2010).**

*Marketing Metrics: The Definitive Guide to Measuring Marketing Performance* (2nd ed.). Pearson Prentice Hall. (A foundational textbook covering CAC, CLV, churn rate, and key performance ratios).

5. **Gupta, S., & Lehmannj, D. R. (2005).**

*Managing Customers as Investments: The Strategic Value of Customers in the Long Run*. Wharton School Publishing. (Extensive academic research into Customer Lifetime Value and retention strategies).

6. Ash, T., Page, R., & Ginty, M. (2012).

*Landing Page Optimization: The Definitive Guide to Testing and Tuning for Conversions* (2nd ed.).

Wiley. (Details conversion rate mechanics, web usability, and user testing methodology).

7. Blattberg, R. C., Kim, B. D., & Neslin,

S. A. (2008). *Database Marketing:*

*Analyze and Manage Customers.*

Springer Science & Business Media.

(Covers incremental revenue modeling, baseline sales estimation, and campaign measurement).

8. Farris, P. W., Bendle, N. T., Pfeifer, P.

E., & Reibstein, D. J. (2010).

*Marketing Metrics: The Definitive*

*Guide to Measuring Marketing*

*Performance* (2nd ed.). Pearson

Prentice Hall. (Provides standardized formulas for ROI, campaign measurement, and financial

evaluation)

9. Kotler, P., & Keller, K. L. (2016). *Marketing Management* (15th ed.). Pearson. (Provides foundational frameworks for designing marketing campaigns and conducting post-campaign evaluations).

10. Farris, P. W., Bendle, N. T., Pfeifer, P. E., & Reibstein, D. J. (2010).

*Marketing Metrics: The Definitive Guide to Measuring Marketing Performance* (2nd ed.). Pearson Prentice Hall. (Covers Customer Retention Rate, CAC, and Average Order Value).

11. Reichheld, F. F. (2003). *The One Number You Need to Grow*. Harvard Business Review, 81(12), 46-54. (The original research article that introduced the Net Promoter Score framework).

12. Kotler, P., & Keller, K. L. (2016).

*Marketing Management (15th ed.).*  
*Pearson. (Provides detailed*  
*frameworks on customer journeys,*  
*buying behaviors, and sales funnel*  
*structures).*

13. Ash, T., Page, R., & Ginty, M. (2012).

*Landing Page Optimization: The*  
*Definitive Guide to Testing and*  
*Tuning for Conversions (2nd ed.).*  
*Wiley. (Covers A/B testing*  
*mechanics, bounce rate reduction,*  
*and conversion elements).*

14. Farris, P. W., Bendle, N. T., Pfeifer, P.

*E., & Reibstein, D. J. (2010).*  
*Marketing Metrics: The Definitive*  
*Guide to Measuring Marketing*  
*Performance (2nd ed.). Pearson*  
*Prentice Hall. (Defines formal*  
*calculations for ROAS, CTR, and*  
*digital marketing performance).*

15. Kohavi, R., Tang, D., & Xu, Y. (2020). *Trustworthy Online Controlled Experiments: A Practical Guide to A/B Testing*. Cambridge University Press. (The authoritative textbook on running digital A/B tests and statistical experiments).
16. **Kotler, P., & Keller, K. L. (2016).** *Marketing Management (15th ed.)*. Pearson. Covers core principles of lead capture, customer acquisition funnels, and marketing effectiveness.
17. **Optimizely Marketing Glossary.**  
"What is Lead Generation?" Defines how digital businesses capture customer interest via forms, content, and touchpoints.

18. **Forbes Business Council (2022).** *"A Guide To Marketing Attribution."* Breaks down attribution models (first-click, last-click, multi-touch) in business applications.
19. **Zendesk Sales & Marketing Guide.** *"What is lead generation?"* Explains the distinction between raw leads, qualified leads, and inbound/outbound strategies.